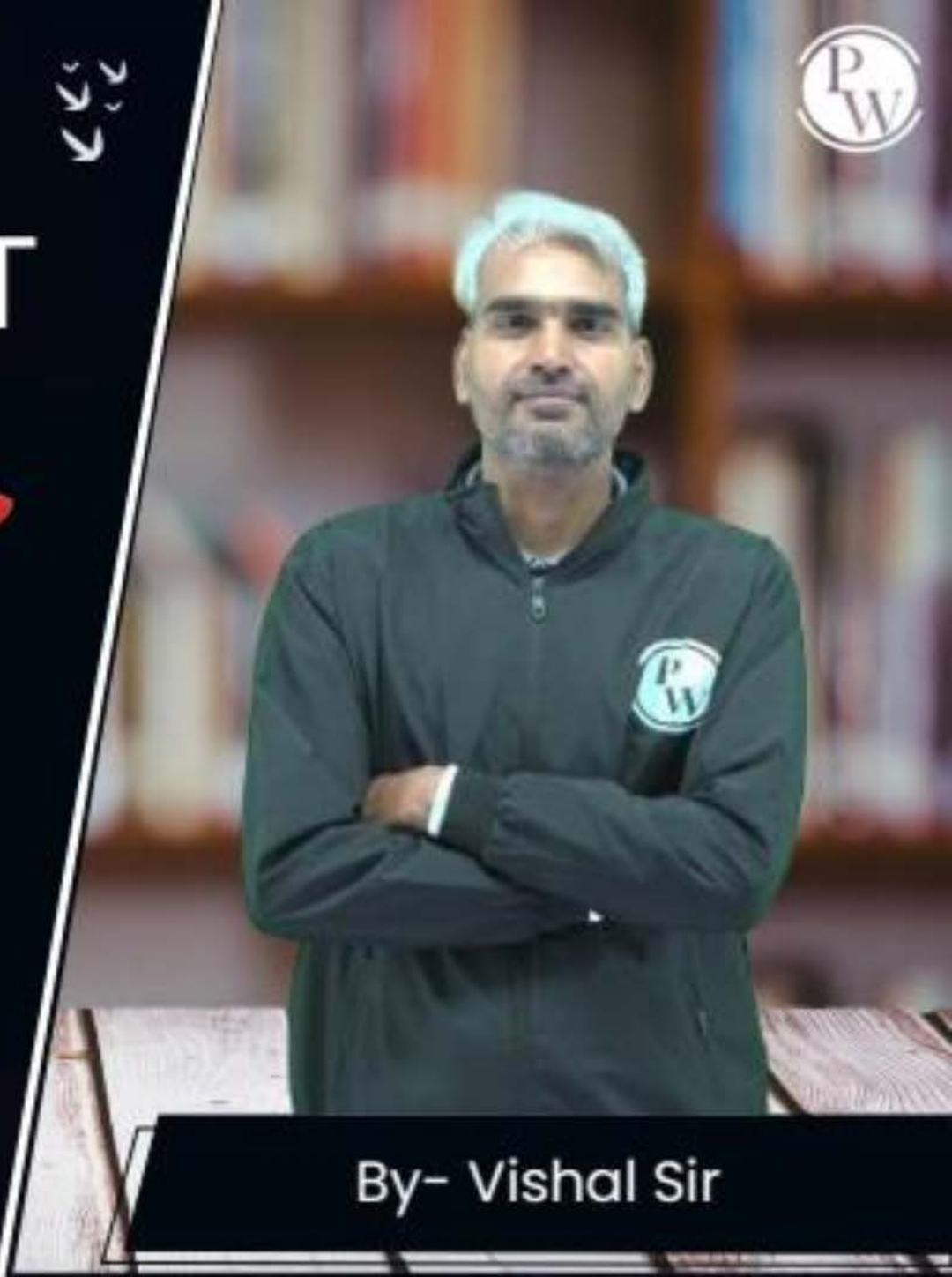
Computer Science & IT



Database Management System

Query Languages

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



- Topic: SQL
- Topic: SQL clauses
- Topic: Aggregate functions

Slide

Topics to be Covered



- Topic SQL commands
- Topic SQL clauses

Slide



Topic : SQL

Prerequisite :- Relational Algebra

Structured Query language

It is a non-procedural query language

i.e. we must understand the "Syntax" of SQL



Topic : Commands in SQL

Commands in SQL

- Data definition Language (DDL)
  - Create
  - Drop
  - Alter
- Data Manipulation language (DML)
  - Insert
  - Delete
  - Update
  - Truncate
- Data Control language (DCL)
  - Grant
  - Revoke
- Transaction Control Language (TCL)
  - Commit
  - Rollback/Abort
- Data query language (DQL)
  - Select



Topic : SQL clauses

1. From
2. Where
3. Group by
4. Having
5. Order by

Basic Syntax of SQL query :

It is optional
If we do not use "distinct" then SQL query output may contain duplicate tuples

$\pi_{A_1, A_2, \ldots A_K}$ ≡ ① Select distinct $A_1, A_2, \ldots, A_K$

List of attributes required in output

$R_1 \times R_2 \times \ldots \times R_m$ ≡ ② From $R_1, R_2, \ldots, R_m$

List of database tables required for the execution

$\sigma_{(\text{Cond}^n \text{ to select tuples})}$ ≡ ③ Where (Condition to select tuples)

No Equivalent R.A. Expression

④ Group by (Attribute/s)

⑤ Having (Cond^n to select groups)

⑥ Order by (Attributes) DESC / ASC

Projection (π) in R.A. $\not\equiv$ Select in SQL

because duplicate may be present in SQL

Projection (π) in R.A. $\equiv$ Select distinct in SQL



Topic : Aggregate functions

There are five different aggregate functions in SQL

1. Count ( )
2. Sum ( )
3. Avg ( )
4. Max ( )
5. Min ( )



Topic : Aggregate functions

- Select Count (*)
  From STUDENT=
- Select Count (Sid)
  From STUDENT =
- Select Count (distinct Sid)
  from STUDENT=
- Select Count (Marks)
  from STUDENT=
- Select Count (distinct Marks)
  From STUDENT=

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

Everything

- Count(*) - It returns the total number of tuples in the relation. Count(*) will also count the empty tuples, i.e., the tuples in which all values all NULL.

- Count(Attribute) - It will count the total number of non NULL values in the column corresponding to the attribute specified with count function.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Count (*) From STUDENT = 7
- Select Count (Sid) From STUDENT = 6
- Select Count (distinct Sid) from STUDENT = 6
- Select Count (Marks) From STUDENT = 5
- Select Count (distinct Marks) From STUDENT = 4

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : NOTE

- NULL is a non-zero unknown value.
- No two NULL are equal. {∵ Comparison operators are not used to compare two NULL values}
- NULL values are always discarded by aggregate functions except for count(*).
- Arithmetic operation with NULL will always produce NULL

NULL + 50 = NULL



Topic : Aggregate functions

- Sum(Attribute) - It will return the summation of all non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

- Sum(distinct Attribute) - It will return the summation of all distinct non NULL values in the column corresponding to the attribute specified with aggregate function Sum().

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Sum (Marks) From STUDENT = 250

- Select Sum (distinct Marks) From STUDENT = 190

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Avg(Attribute) - It will return the average of all non NULL values with respect to specified attribute.

$$\text{Avg (Attribute)} = \frac{\text{Sum (Attribute)}}{\text{Count (Attribute)}}$$

- Avg(distinct Attribute) - It will return average of all distinct non NULL values with respect to specified attribute.

$$\text{Avg (distinct attribute)} = \frac{\text{Sum (distinct attribute)}}{\text{Count (distinct attribute)}}$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Avg (Marks) From STUDENT = (50)

$$= \frac{\text{Sum (Marks)}}{\text{Count (Marks)}} = \frac{250}{5} = 50$$

- Select Avg (distinct Marks) From STUDENT:

$$= \frac{\text{Sum (distinct Marks)}}{\text{Count (distinct Marks)}} = \frac{190}{4} = 47.5$$

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Max(Attribute) - It will return the maximum of all the values in the column corresponding to the specified attribute.

- Min(Attribute) - It will return the minimum of all the values in the column corresponding to the specified attribute.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Aggregate functions

- Select Max (Marks) From STUDENT = 70

- Select Min (Marks) From STUDENT = 20

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	B	60	IT
S3	A	20	CS
S4	C	NULL	IT
S5	B	60	EC
S6	D	70	EC
NULL	NULL	NULL	NULL



Topic : Commands in SQL

- CREATE TABLE - creates a new table
- ALTER TABLE - modifies a table → { used to add/delete Column in database table }
- DROP TABLE - deletes a table
  - It will delete Complete schema
- INSERT - inserts new data into a database table (row/rows)
- DELETE - deletes data from a database table (row/rows)
- UPDATE - updates data in a database

- SELECT - extracts data from a database



Topic : Commands in SQL

- CREATE TABLE - creates a new table

```
CREATE TABLE Student (
        Sid varchar(25),
        Sname varchar(25),
        Marks int,
                        );
```

```
Create table
    New_table_name
     ( Att1   data-type   Constraint,
       Att2   data-type   Constraint,
     );
```

optional → Constraint



Topic : Commands in SQL

- CREATE TABLE - creates a new table

```
CREATE TABLE Student (
        Sid varchar(25),
        Sname varchar(25),
        Marks int,
                );
```

⇒

Student

Sid	Sname	Marks



Topic : Commands in SQL

- ALTER TABLE - modifies a table

Add new Column (or) Delete existing Column

name_of_table

Constraint

```
ALTER TABLE Student
ADD Branch varchar(25);
```

New attribute name

data-type

Student

Sid	Sname	Marks

* To add a new Column we use ADD
* To delete an existing Column we use DROP



Topic : Commands in SQL

- ALTER TABLE - modifies a table

ALTER TABLE *Student*
ADD *Branch varchar(25);* ⟹

Student

Sid	Sname	Marks	Branch

- If we write

Alter table Student
Drop Sname

} As a result, Column Sname will be deleted from Student table.



Topic : Commands in SQL

- DROP TABLE - deletes a table

DROP TABLE *Student*; ⇒ It will delete the complete table along with its structure (schema)

- TRUNCATE TABLE - deletes complete data from the table without deleting the structure of the table

TRUNCATE TABLE Student;



Topic : Commands in SQL

- INSERT - inserts new data(row/rows) into a database table.

- INSERT INTO *Student* VALUES (*S1, A, 35, CS*);

Student

Sid	Sname	Marks	Branch

We can insert multiple tuples using single "insert" Command

```
Insert into table_name
( Values ( Att1, Att2, ...., Attk),
  Values ( Att1, Att2, ...., Attk),
  :  :  :
```



Topic : Commands in SQL

- INSERT - inserts new data(row/rows) into a database table.

- INSERT INTO *Student* VALUES (*S1*, *A*, *35*, CS);

⇒

Student

Sid	Sname	Marks	Branch
S1	A	35	CS



Topic : Commands in SQL

- DELETE - deletes data(row/rows) from a database table

eg: Delete from Student
Where (Marks = 35);

All the tuple from Student table in which Marks = 35 will be deleted



Topic : Commands in SQL

- UPDATE - updates data in a database table

```
UPDATE Student
SET Marks = Marks + 5;
```

Student

Sid	Sname	Marks	Branch
S1	A	35	CS

```
Update    table_name
Set       Updation required
```



Topic : Commands in SQL

- UPDATE - updates data in a database

UPDATE *Student*
SET *Marks = Marks + 5;*

Student

Sid	Sname	Marks	Branch
S1	A	35+5=40	CS



Topic : SQL clauses

1. •FROM - The FROM clause in SQL is used to select the database tables

2. •WHERE - The WHERE clause in SQL is used to select the tuples from the database table based on conditions specified with WHERE clause. { If "Where" Clause is not present, then all rows will be selected }

3. •GROUP BY - GROUP BY clause is used to group the result of WHERE clause based on attributes specified with GROUP BY clause. { If Where Clause is not Present, then Group by Clause will be applied on all rows } { If Group by Clause is not present, then we will assume that all tuples togethe form a single group }

4. •HAVING - HAVING clause can be used with or without GROUP BY clause. If group by clause is present, then condition specified with having clause will be used to select the groups that satisfy the specified condition. { Having Condn is applied on each group } { If Having Clause is not present then all the groups will get selected }

•ORDER BY - The ORDER BY clause in SQL is used for sorting the records of the database based on attributes specified with ORDER BY clause.



Topic : Order of execution

Order of Execution:-

1. From
2. Where
3. Group By
4. Having
5. Select
6. Order BY



Topic : SQL clauses

- FROM:- From clause is used to select the tables from the database.

Everything

```
Select *
From Student
```

⇓

O/p of this query will be Complete Student table

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

- FROM:- From clause is used to select the tables from the database.

- Select Sid
  From Student ⇒ O/P :

Sid
S1
S2
S3
S4
S5
S6

- Select distinct Sid
  From Student = O/P

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

- FROM:- From clause is used to select the tables from the database.

- Select Sname From Student ⇒ o/p :

Sname
A
A
B
A
C
C

- Select distinct Sname From Student ⇒ o/p :

Sname
A
B
C

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

* Retriere Sids of students who Scored More than 40 Marks

```
Select Sid
    From Student
        Where Marks > 40
```

⇒ O/P =

Sid
S3
S4

Student

Sid	Sname	Marks	Branch
S1	A	✗ 40	CS
S2	A	✗ 20	IT
S3	B	✓ 60	CS
S4	A	✓ 60	EC
S5	C	✗ 40	IT
S6	C	✗ NULL	EC



Topic : SQL clauses

WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

* Retriere Sids of students where Marks are more than or equal to 20 And less than or equal to 40

```
Select Sid
From Student
Where (Marks >= 20 AND Marks <= 40)
```

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

{If Where clause is not present then all the rows will be grouped based on attributes specified with group by}

Query: Retrieve names of all branches along with maximum marks in that branch.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

* Select Max (Marks)
  From Student

$\Rightarrow$ o/p =

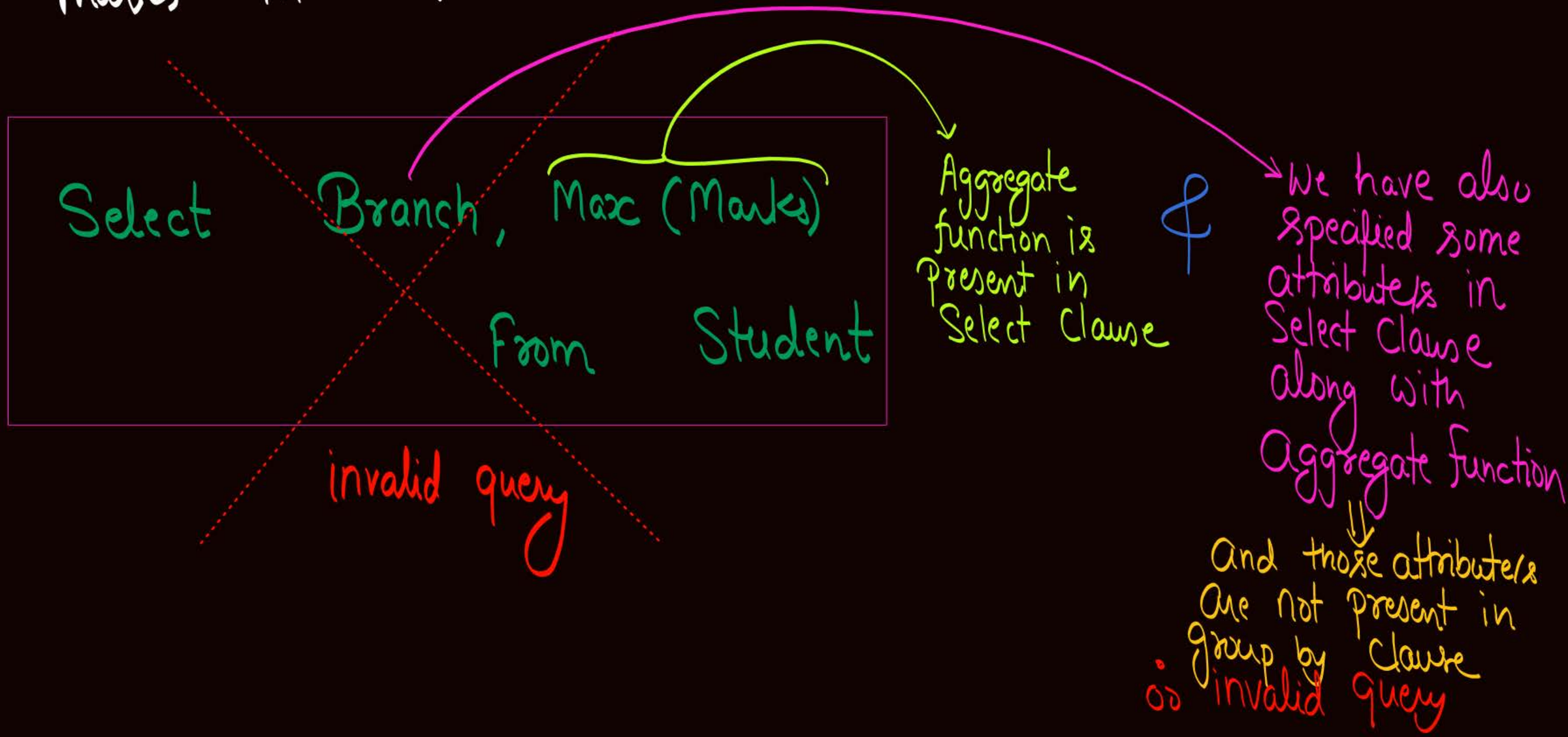
Max(Marks)
60

* Select distinct Branch
  From Student

$\Rightarrow$ o/p =

Branch
CS
IT
EC

→ Retrieve name of all Branches along with Maximum marks in that Branch.



Note:- If "group by" clause is present, then aggregate functions will be applied on each group.



Topic : SQL clauses

→ If attribute is present in Select Clause along with aggregate function, then that attribute must be present in group by clause.

NOTE:-

1. We can not select any attribute in SELECT clause along with aggregate function if those attributes are not present in GROUP BY clause.
   But, If attributes are present in Group by clause, then it is not necessary to select some or all of them using select clause along with aggregate function
2. If aggregate function is used along with GROUP BY clause, then aggregate function is applied on each group.



Topic : SQL clauses

GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

Query: Retrieve names of all branches along with maximum marks in that branch.

```
Select Branch, Max(Marks)
From Student
Group by (Branch)
```

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

O/P :

Branch	Max(Marks)
CS	60
IT	40
EC	60



Topic : SQL clauses

HAVING:- HAVING condition is applied on each group. (Provided Group by clause is present)

Query: Retrieve branch names with average marks more than or equal to 40.

Select Branch ⇒ o/p =

Branch
CS
EC

Select Branch
From Student
Group by (Branch)
Having (Avg(Marks) ≥ 40)

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Avg(Marks)
50
× 30
160



2 mins Summary

Topic SQL commands

Topic SQL clauses

Slide

THANK - YOU